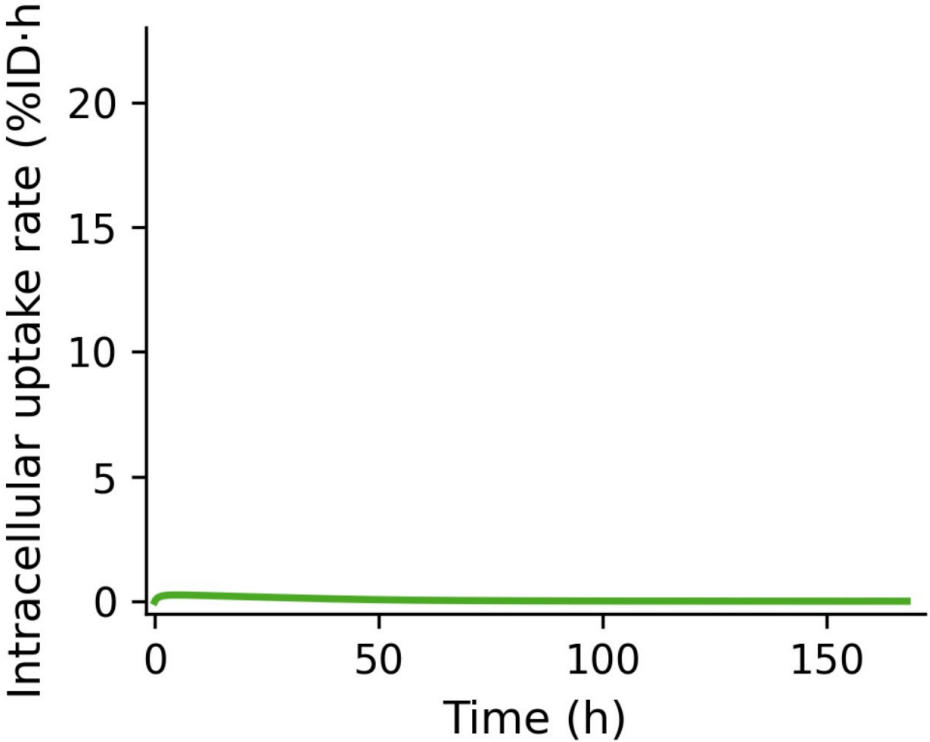
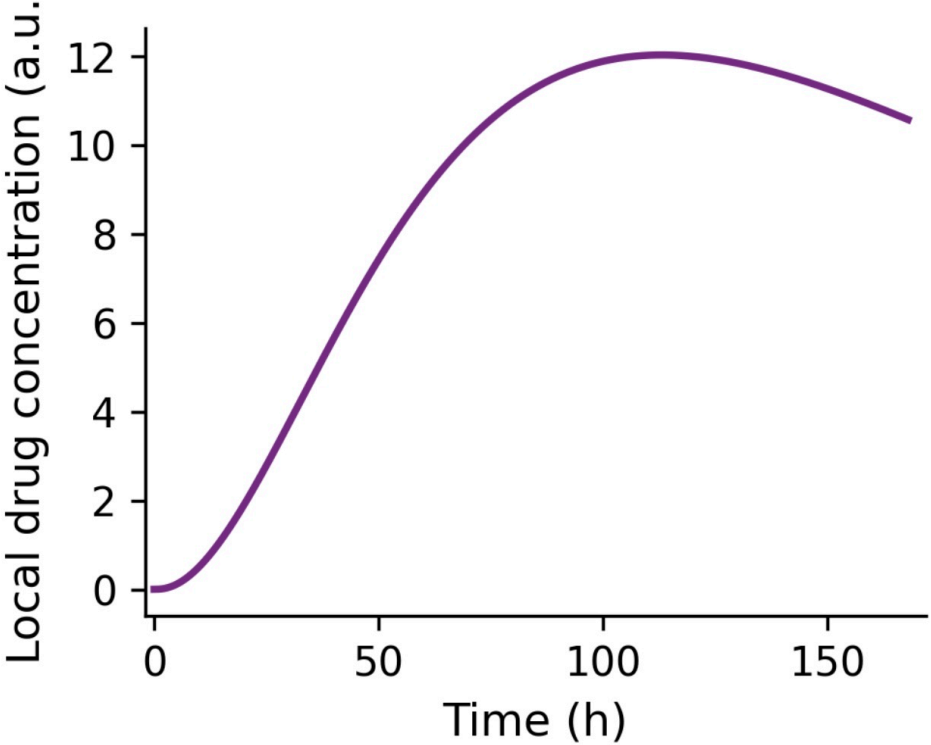
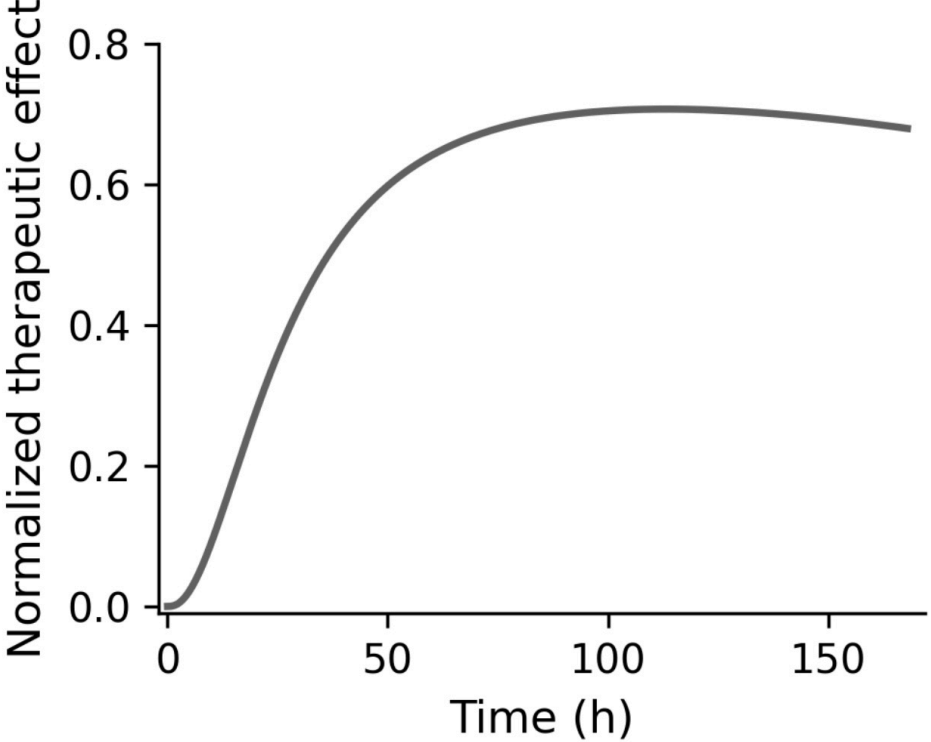
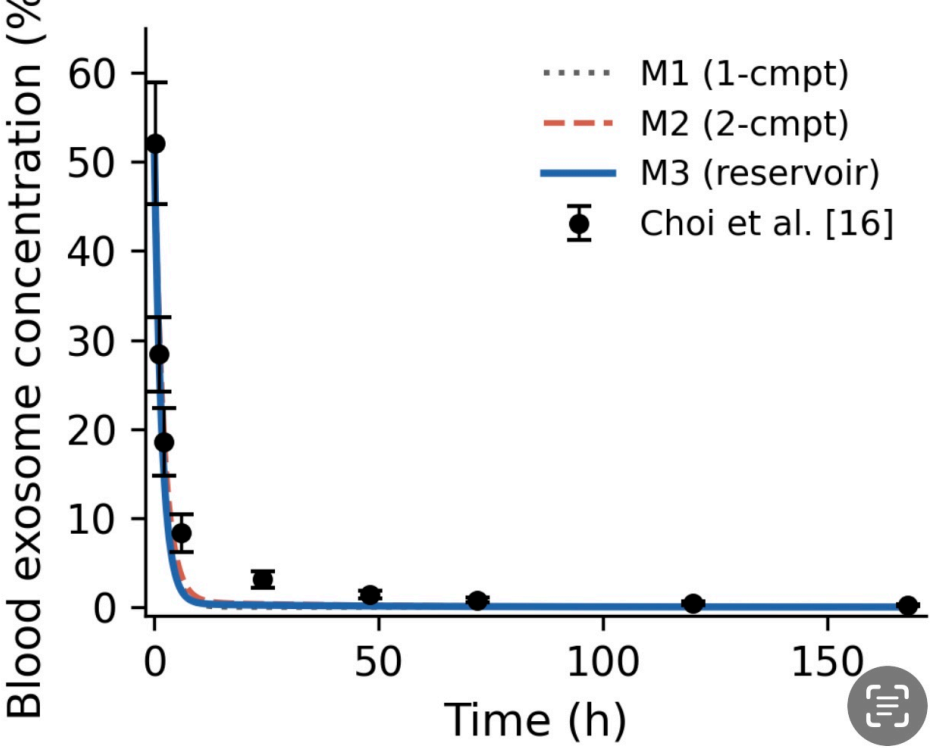
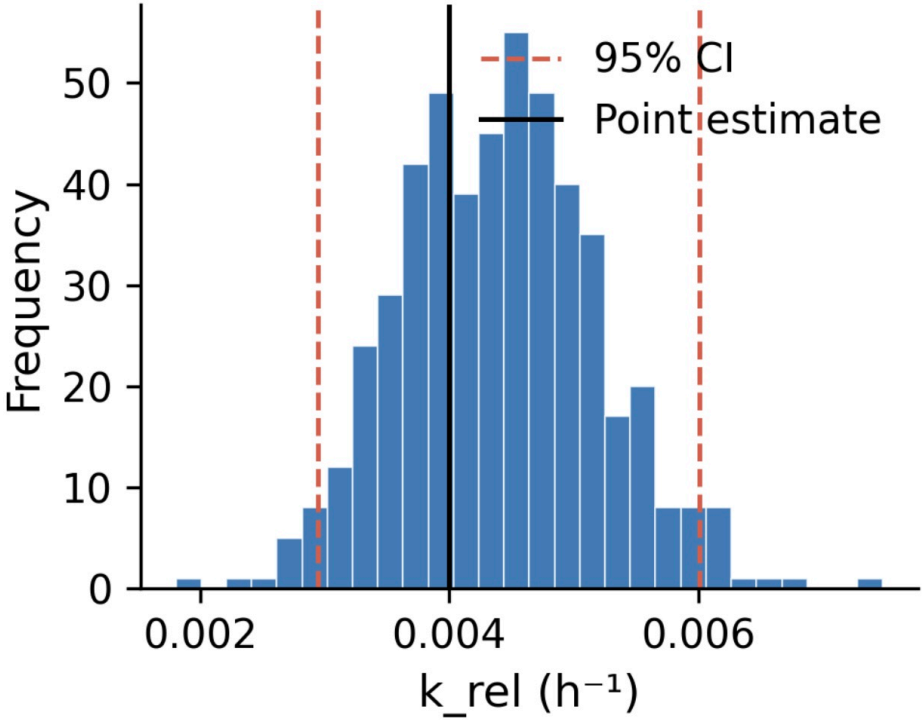
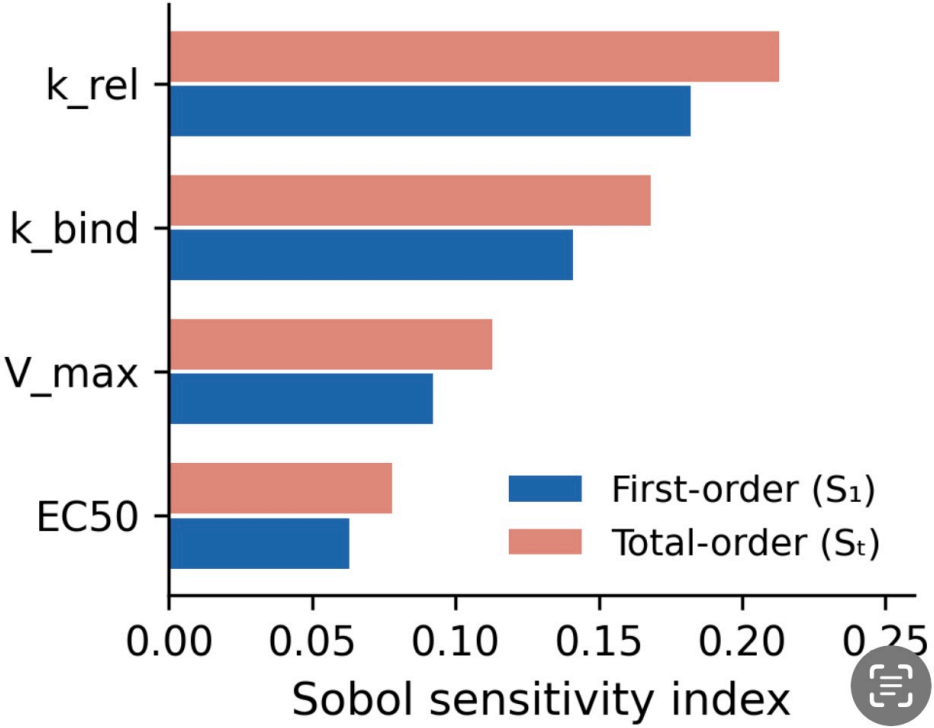

Intracellular uptake rate (%ID·h
20
15
10
5
0
0
50
100
150
Time (h)

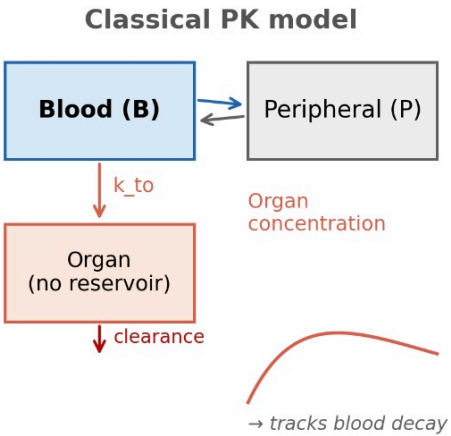

Local drug concentration (a.u.)
12
10
8
6
4
2
0
0
50
100
150
Time (h)

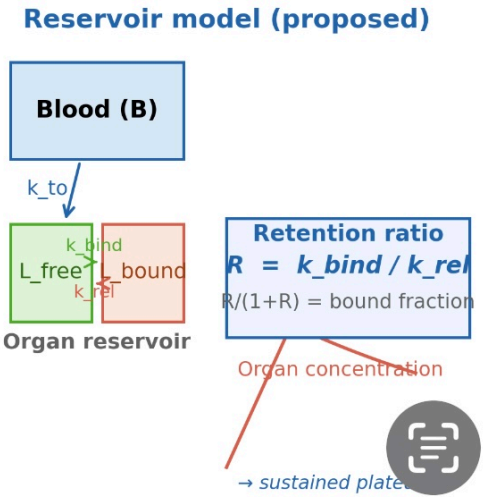

Normalized therapeutic effect
0.0
0.2
0.4
0.6
0.8
0
50
100
150
Time (h)

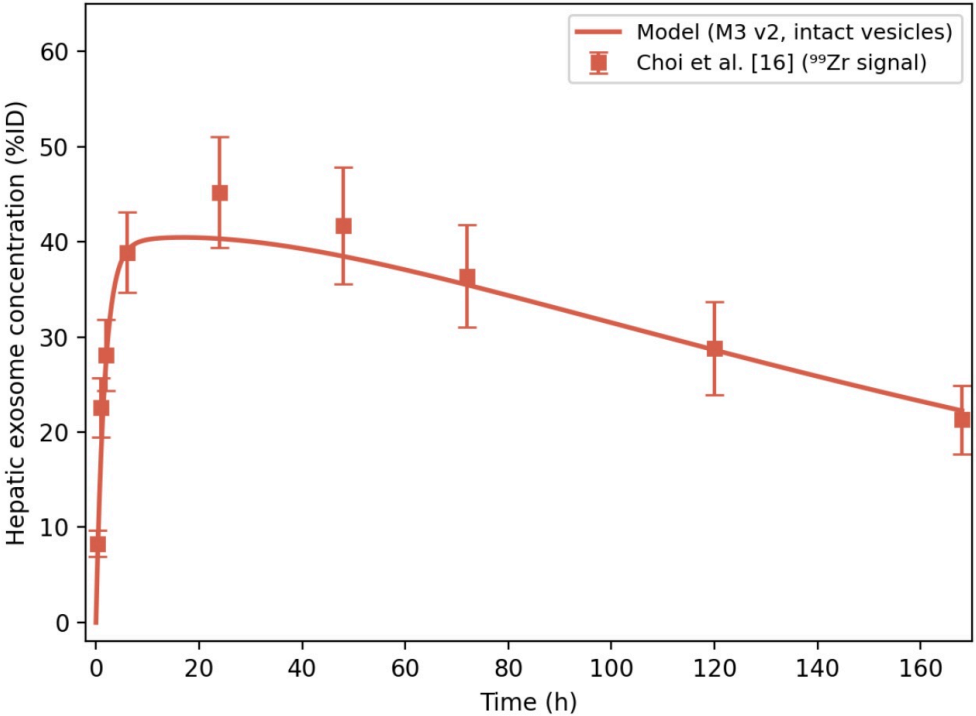

Blood exosome concentration (%
60
50
40
30
20
10
0
0
50
100
150
Time (h)
M1 (1-cmpt)
M2 (2-cmpt)
M3 (reservoir)
Choi et al. [16]

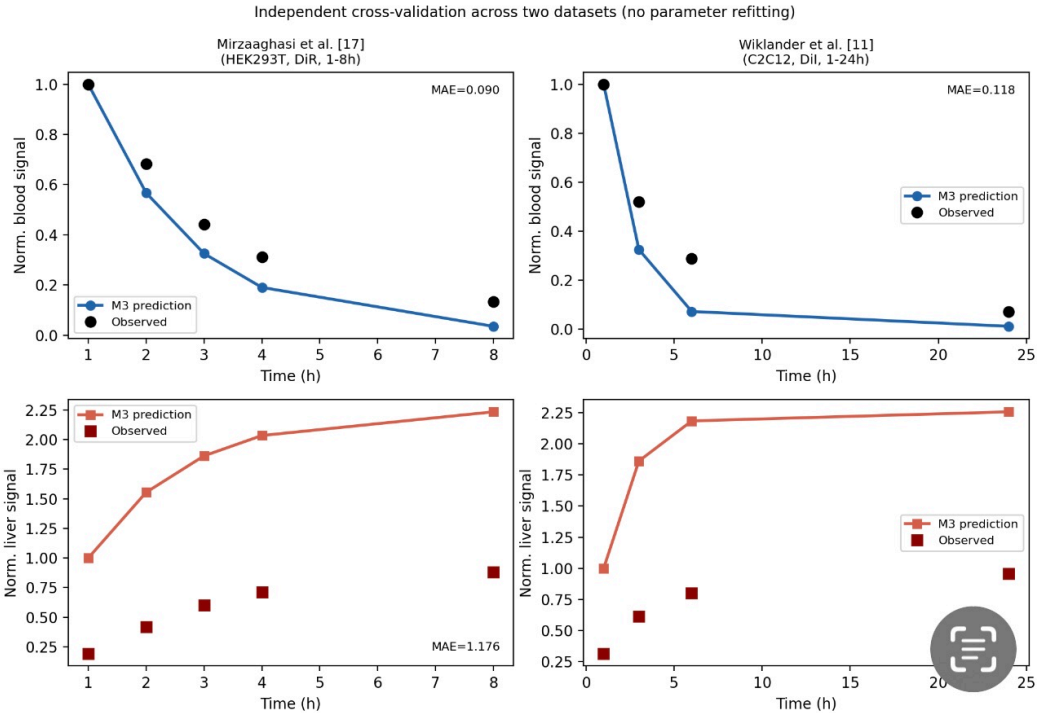

95% CI
Point estimate
Frequency
0
10
20
30
40
50
0.002
0.004
0.006
k_rel (h⁻¹)


k_rel
k_bind
V_max
EC50
First-order (S_1)
Total-order (S_t)
0.00
0.05
0.10
0.15
0.20
0.25
Sobol sensitivity index


Classical PK model
Blood (B)
Peripheral (P)
k_to
Organ concentration
Organ (no reservoir)
clearance
→ tracks blood decay


Reservoir model (proposed)
Blood (B)
k_to
k_bind
L_free
L_bound
k_rel
Organ reservoir
Retention ratio
R = k_bind / k_rel
R/(1+R) = bound fraction
Organ concentration
→ sustained plate


Model (M3 v2, intact vesicles)
Choi et al. [16] (⁹⁹Zr signal)
Hepatic exosome concentration (%ID)
Time (h)
0
10
20
30
40
50
60
0
20
40
60
80
100
120
140
160


Independent cross-validation across two datasets (no parameter refitting)
Mirzaaghasi et al. [17]
(HEK293T, DiR, 1-8h)
Wiklander et al. [11]
(C2C12, DiI, 1-24h)
MAE=0.090
MAE=0.118
MAE=1.176
Norm. blood signal
Norm. liver signal
Time (h)
M3 prediction
Observed

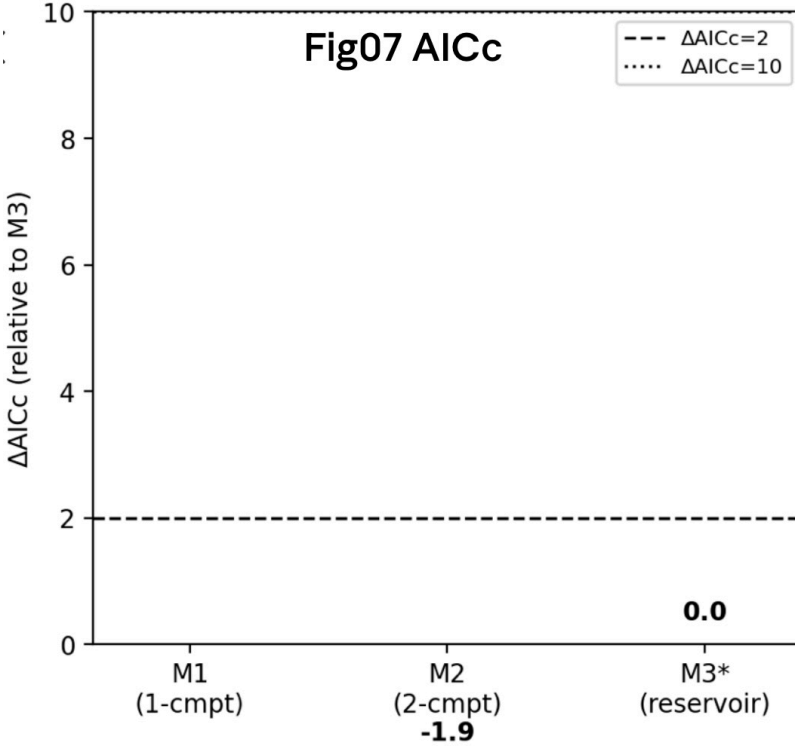
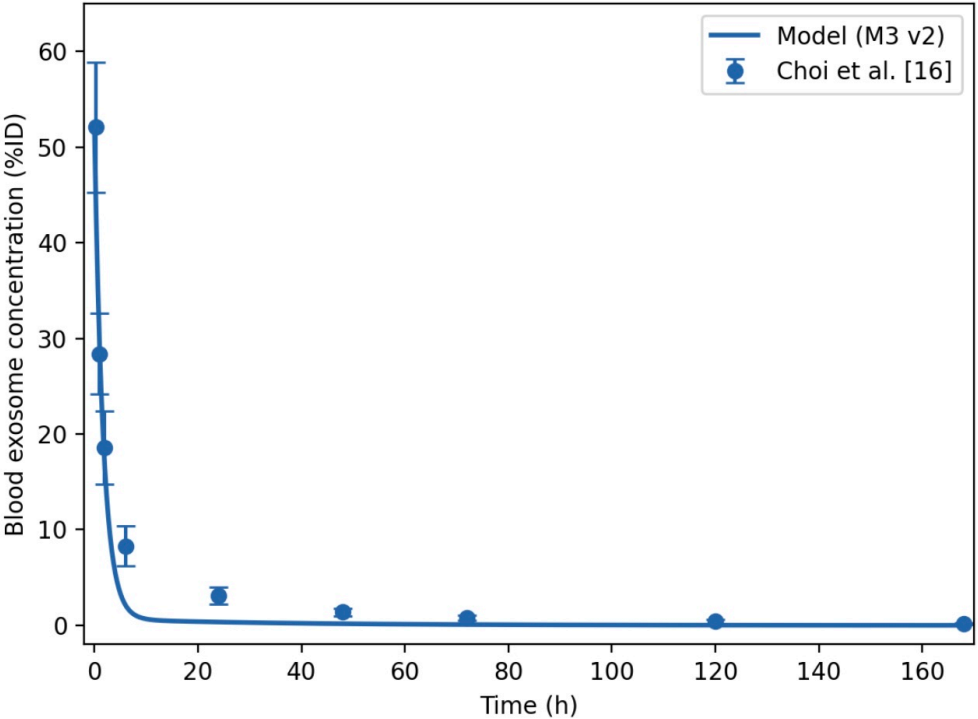
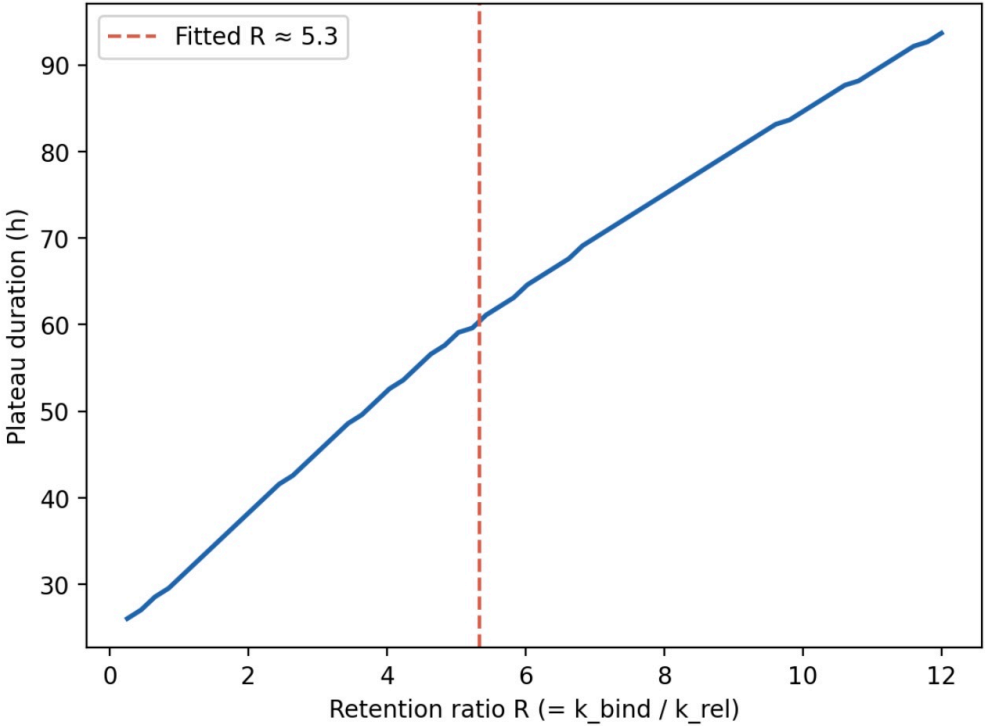
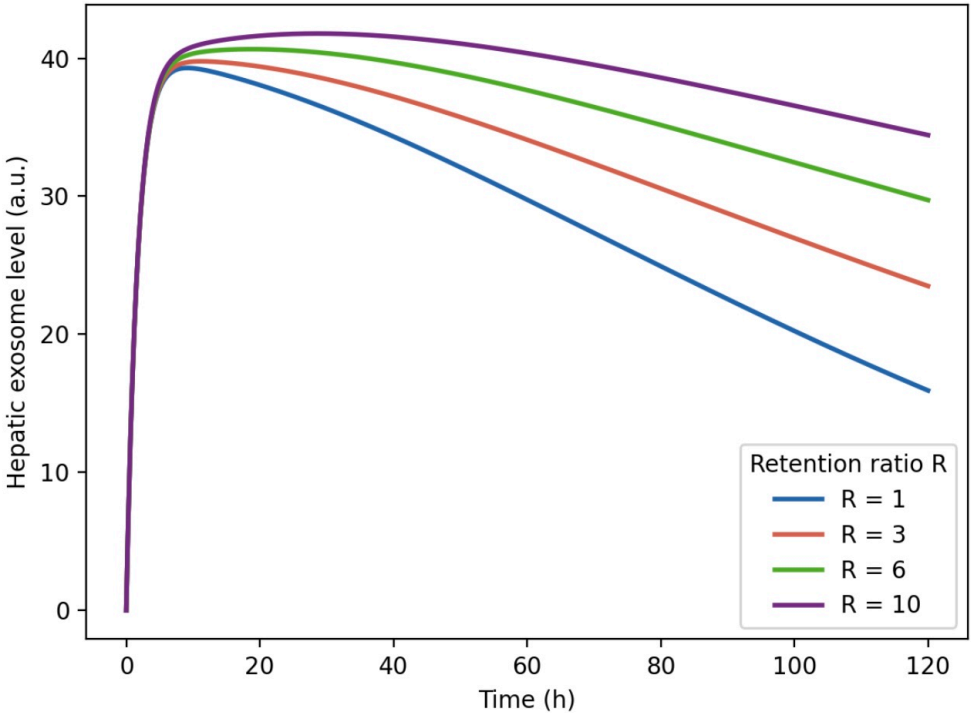
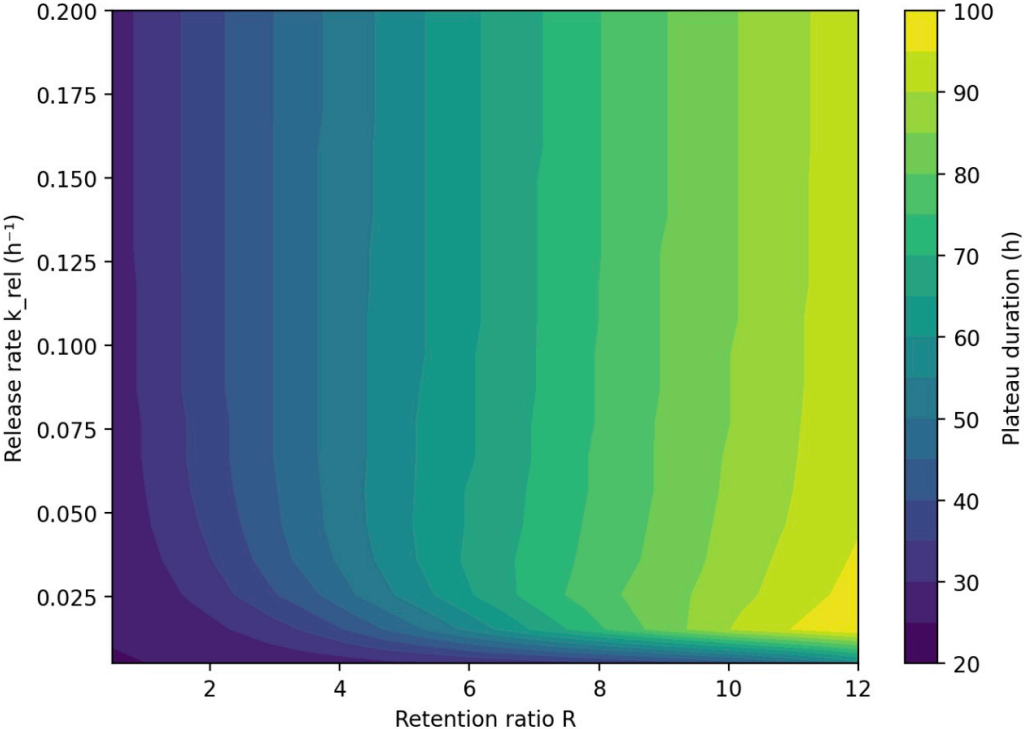


Fig07 AICc
ΔAICc=2
ΔAICc=10
ΔAICc (relative to M3)
10
8
6
4
2
0
0.0
M1
(1-cmpt)
M2
(2-cmpt)
-1.9
M3*
(reservoir)


Model (M3 v2)
Choi et al. [16]
Blood exosome concentration (%ID)
Time (h)
0
10
20
30
40
50
60
0
20
40
60
80
100
120
140
160


Fitted R ≈ 5.3
Plateau duration (h)
Retention ratio R (= k_bind / k_rel)
30
40
50
60
70
80
90
0
2
4
6
8
10
12


Hepatic exosome level (a.u.)
0
10
20
30
40
0
20
40
60
80
100
120
Time (h)
Retention ratio R
R = 1
R = 3
R = 6
R = 10


0.200
0.175
0.150
0.125
0.100
0.075
0.050
0.025
Release rate k_rel (h⁻¹)
2
4
6
8
10
12
Retention ratio R
100
90
80
70
60
50
40
30
20
Plateau duration (h)